Q1 - 29 January - Shift 1

In a metre bridge experiment the balance point in obtained if the gaps are closed by 2Ω and 3Ω. A shunt of $X\Omega$ is added to 3Ω resistor to shift the balancing point by 22.5 cm. The value of X is ____

Space for your notes:

Q2 - 29 January - Shift 2

A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω. When a resistance of 15 Ω is used for shunting null point moves to 300 cm. The internal resistance of the cell is ______ Ω.

Space for your notes:

Q3 - 30 January - Shift 1

In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by 0.5 mm on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46^{th} division the circular scale coincide with the reference line. The diameter of the wire is ______ $\times 10^{-2}$ mm.

Space for your notes:

Q4 - 01 February - Shift 1

In an experiment to find emf of a cell using potentiometer, the length of null point for a cell of emf 1.5 V is found to be 60 cm. If this cell is replaced by another cell of emf E. the length-of null point increases by 40 cm. The value of E is $\frac{x}{10}$ V. The value of x is ___________.

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2) **Q2** (5) **Q3** (22) **Q4** (25)

Q1 (2)

$$\frac{2}{\left(\frac{3x}{3+x}\right)} = \frac{40+22.5}{60-22.5} = \frac{62.5}{37.5} = \frac{5}{3}$$

$$\frac{6}{5} = \frac{3x}{3+x}$$

$$6+2x = 5x \Rightarrow x = 2$$

Q2 (5)

$$\frac{\varepsilon}{r+5} \times 5 = 200x \qquad \ldots(1)$$

$$\frac{\varepsilon \times 15}{r+15} = 300x \qquad \ldots(2)$$

$$\Rightarrow r = 5\ \Omega$$

Ans. 5

Q3 (22)

$$\text{Least count} = \frac{\text{Pitch}}{\text{No. of circular divisions}}$$

$$= \frac{0.5\text{ mm}}{100}$$

Least count $= 5 \times 10^{-3}$ mm

Positive Error = MSR + CSR (LC)

$= 0 \text{ mm} + 6\,(5 \times 10^{-3} \text{ mm})$

Reading of Diameter = MSR + CSR (LC)

Positive zero error

$= 4 \times 0.5 \text{ mm} + (46(5 \times 10^{-3})) - 6(5 \times 10^{-3})$ mm

Q4 (25)

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{1.5}{E_2} = \frac{60}{60+40} = \frac{6}{10} = \frac{3}{5}$$

$$E_2 = \frac{5}{2} = \frac{x}{10}$$

$x = 25$